
Theorem 1. If X and Y are random variables, then the joint pdf of $X + Y$ is given by convolution:

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} f_X(z)f(t-z)dz$$

Definition 2. The uniform distribution on $[a, b]$ is

$$U(a, b) = \frac{1}{b-a}\mathbf{1}_{[a,b]}$$